

Staples STEM Journal

• AI modeling and complex protein structures • New frontiers in solar science and sunspot prediction • Medical diagnostics and artificial intelligence • Artificial intelligence and RNA viruses • GIMPS and the search for new prime numbers • Wood burial and carbon removal • The impact of K2P channels on health • CRISPR technology's use in medicine •

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***The Staples STEM Journal* provides an outlet for individuals to share their STEM interests with the Staples High School community, and aims to broaden public interest and knowledge in these fields.**

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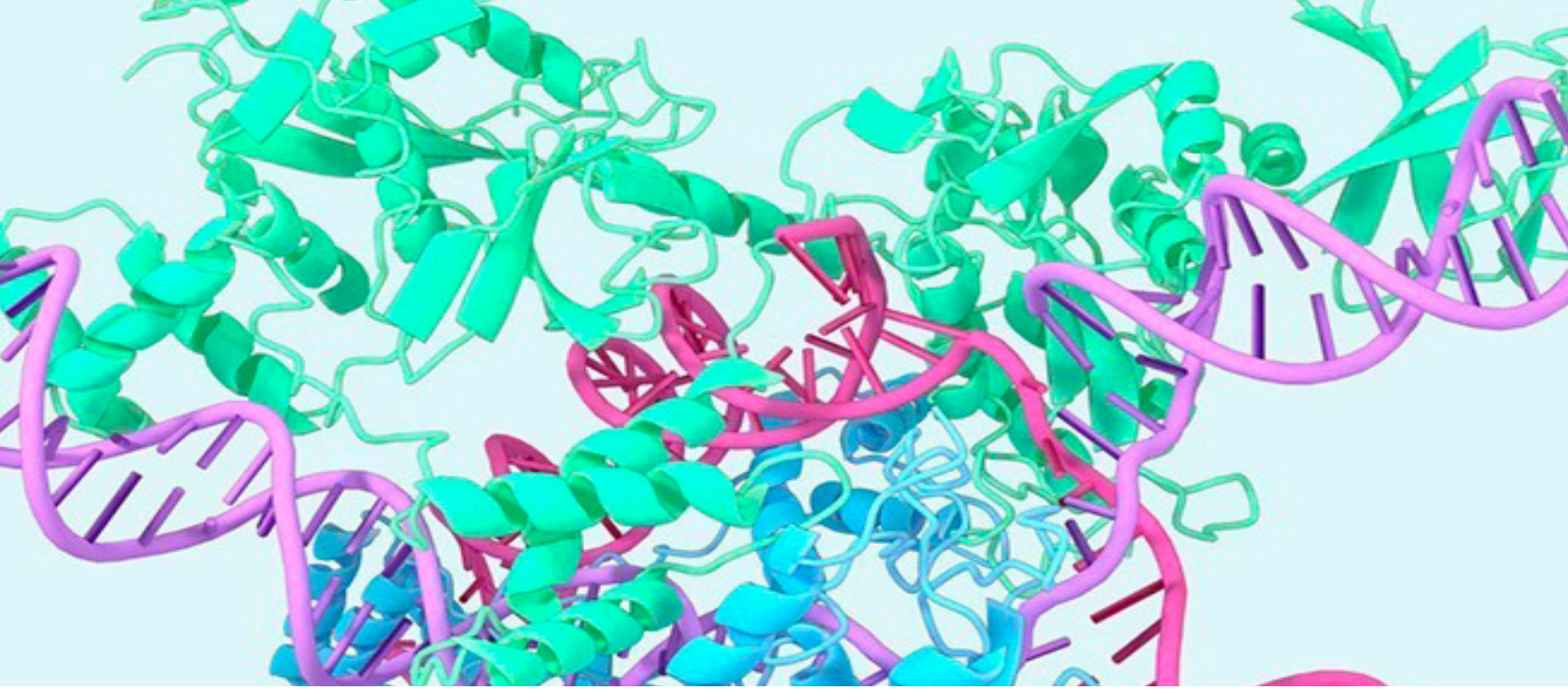
Letter from the Editor



Dear Reader,

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Sincerely,
Samuel Zwick-Lavinsky, ‘25
Editor-in-Chief, Staples STEM Journal



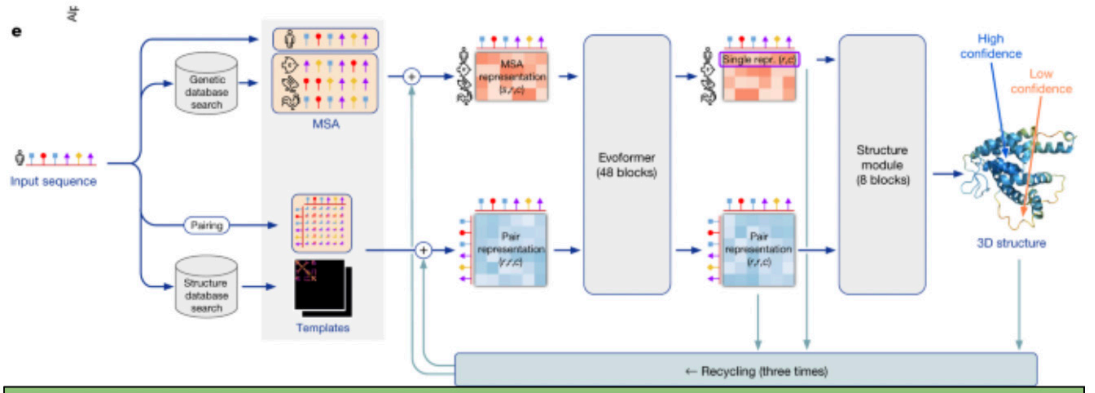
New AI model is now able to generate a 3D model of complex protein structures

Luke Cooper, '27

On May 8th, 2024, Google's DeepMind team finally released AlphaFold3. AlphaFold3 is the newest model of a collection of models created by Google's DeepMind team to help combat the problem of mapping out protein structures. Previous models of AlphaFold were only able to predict very simple proteins. However, AlphaFold3 is able to predict the 3D structures of complex proteins. This includes proteins with nucleic acids, small molecules, and chemical modifications. As a result of this, AlphaFold3 is able to predict almost any type of protein structure under any circumstances, which allows it to be used in the real world to help with drug discovery and our understanding of biology. AlphaFold3 has already been used over 1000 times in scientific research paper publications meaning that AlphaFold3 has already helped scientists 1000 times to make new discoveries. This article will examine how AlphaFold3 works and how it is affecting the world now a few months after its initial release.

Why we need AlphaFold3

Predicting protein shapes is essential in biology because the shape of a protein largely determines what it does in the body. Sickle Cell Anemia is a genetic disease that affects the structure of your blood vessels. This occurs because of one wrong nucleotide in the person's DNA. This wrong nucleotide messes up the amino acid sequence of the protein and ends up messing up the protein's structure. As a result of this, people with sickle cell anemia often face blood clots because the weirdly shaped blood cells often get stuck to the sides of the blood vessels. This creates clumps and blockages in the person's blood vessels. Using AlphaFold3, you would be able to completely map out the structure of the sickle cell protein in only a few minutes compared to the years it usually takes from X-ray Crystallography. X-ray Crystallography is a process of crystallizing the proteins and then shooting them with X-rays. Then the diffraction pattern of the X-rays can be used to find the structure of the protein. This process generates accurate results but it takes a long time. By find-



Structure of the training model for Alpha Fold3. (Jumper, Nature)

-ing the structure of the protein. This process generates accurate results but it takes a long time. By finding the structure of the protein fast it helps researchers create a way to help the patient to get over the disease, maybe even saving a few lives. Such as when the original AlphaFold helped researchers find the structure of the SARS-CoV-2 virus. By doing this they were able to create a vaccine for Covid-19 in a way that uses the structure of the virus to destroy it since most viruses and proteins act with their environment via their structure. However this was AlphaFold1 and AlphaFold2, so why is AlphaFold3 so much better? Well, researchers have long wanted to predict how different types of molecules interact together and how this affects the structure of a protein. AlphaFold3 is able to achieve this by accurately predicting interactions within complex structures, like protein-protein, protein-DNA, and even protein-ligand complexes. As Michael Nuñez from Venture Best notes, "AlphaFold 3 represents a quantum leap beyond its predecessors. While AlphaFold 2 could predict protein structures, version 3 can model the complex interactions between proteins, DNA, RNA, and small molecules — the fundamental processes of life."(Nuñez, VentureBest).

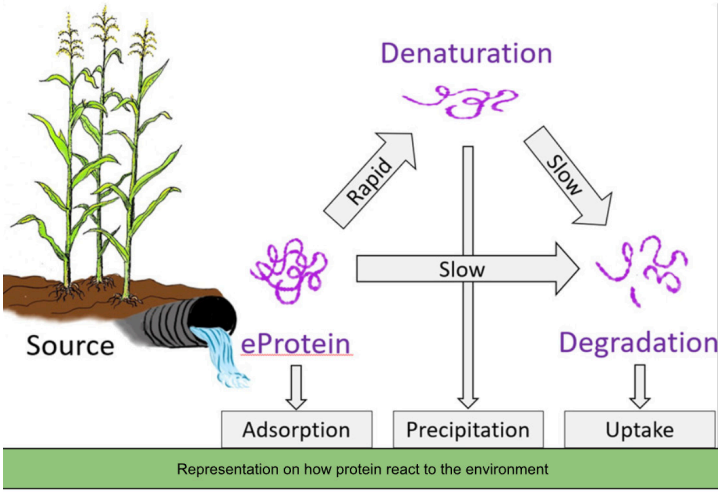
How AlphaFold3 Works

AlphaFold3 was trained like most other AI's, where it is given a large dataset and then trained off of it. What makes it different though is the dataset and the way they trained the model. In the dataset the researchers added already known protein structures, nucleotide sequences and most importantly molecular interactions. This allows AlphaFold3 to predict the way molecules and nucleotides interact with each other to create a protein. The model uses something called a "pairformer" architecture. This system doesn't depend as much on aligning protein sequences (a method known as multiple sequence alignment, or MSA) as earlier models did. Instead, AlphaFold3 directly predicts the positions of atoms, using a process to gradually build the structure. This method allows it to handle a wider range of interactions. The model does this by starting with "noisy" or jumbled-up versions of protein structures, then learns to refine them to match the correct shape. The new approach allows them to more accurately predict the protein structure.

Results and Performance

AlphaFold3 was a major step up over its predecessor's and other softwares. For example, it did 40% better at predicting protein-ligand interactions then AlphaFold2, it did 30% better in predicting DNA and RNA interaction and AlphaFold3 overall had a 90% accuracy for protein-nucleic acid complexes. As Dunger from Nature explains, "In all but one category, [AlphaFold3] achieves a substantially higher performance than strong methods that specialize in just the given task"(Dunger, Nature). As a test to see how powerful AlphaFold3 is, researchers compared it to PoseBusters and the

CASP15 RNA. Both of these programs are leading programs in the protein and RNA structure field. AlphaFold3 easily outperformed both of them with tests showing that AlphaFold3's predictions were more accurate. For example, on tests involving protein-antibody interactions, AlphaFold3 was 35% more accurate than AutoDock Vina concerning predicted ligand positions. All of this shows the improvement AlphaFold3 had from previous protein modeling tools.



Limitations of AlphaFold3

While AlphaFold3 represents a major advancement, it does have some major challenges. For one, the model sometimes produces incorrect or hallucinated structures, as Abramson explains: "generative models are prone to hallucination…where the model may invent plausible-looking structure even in unstructured regions"(Abramson, Nature). This issue can make the results less reliable. Another limitation is that AlphaFold3 can only capture a static picture of proteins. This means that it can not show how the protein will react in a real environment. Also, AlphaFold3 is not yet allowed to be used in official drug discovery. Calloway from Nature describes, "A key limitation of these models is that, like AlphaFold3, none is licensed for commercial applications such as drug discovery"(Calloway, Nature). However, this just means that there is still room for improvement in the field of protein modeling. As of right now Google's DeepMind team has teamed up with Isomorphic Labs so they can make an AlphaFold that can deal with more types of biomolecules, other then just proteins.

The Future of AlphaFold3

AlphaFold3 marks an exciting step forward in not only biology but also AI. It brings a new level of accuracy and versatility to protein modeling, expanding what we can predict from simple protein structures to complex molecular assemblies. As Abramson from Nature summarizes, "AlphaFold3 takes a large step in this direction, demonstrating that it is possible to accurately predict the structure of a wide range of biomolecular systems in a unified framework"(Abramson, Nature). And even though there are still some challenges that AlphaFold3 faces such as hallucinating results, over the future as we see computers and AI get more powerful. It is almost guaranteed that we not only see some of these issues fixed but we also see new creations such as predicting how proteins act in different environments and predicting the structure of other molecular objects.

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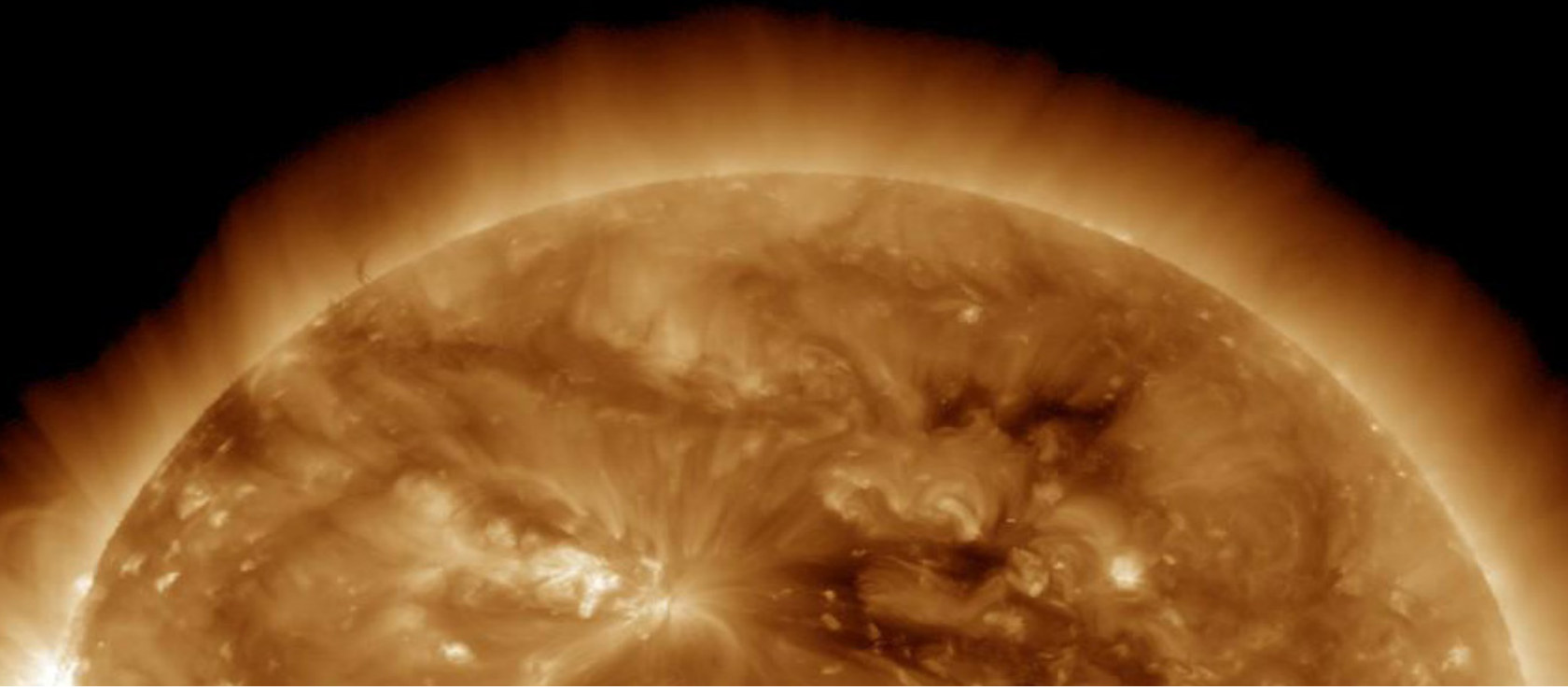
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Solar Tsunamis: Solar Cycle 25 Has Exceeded Expectations and Is Supporting a New Theory of How the Sun Works

Jonah Frey, '28

Solar cycles are both constant and erratic: they wax and wane on predictable timescales, but their intensities are highly variable. An extreme solar cycle can cause flares or coronal mass ejections (CMEs) that damage satellites or electronics, and fluctuations in solar intensity may impact Earth's climate (Hathaway, 2015). To allow for better preparedness against solar activity, as well as to have a better understanding of the Sun's inner workings, scientists have created models that predict the intensity of solar cycles. However, scientists are still refining their predictive methods, and Solar Cycle 25 shows how inaccurate predictions can be; it is currently around double its predicted strength. The field of solar predictions is evolving rapidly, with new models and methods being worked on and older ones being improved.

What are Solar Cycles?

The Sun naturally goes through periods of differing intensity. Over a period of 22 years, the magnetic poles of the Sun switch from the top of the Sun to the bottom, and then swap back (Hathaway, 2015). This 22-year period, called the Hale cycle, is further divided into two 11-year cycles called solar cycles; during each solar cycle, solar activity increases to reach a maximum just before halfway through the cycle and then goes back down to a minimum by the end of the cycle. During a solar cycle's maximum, the magnetic fields of the Sun cross over one another as they shift in orientation, causing disturbances and odd patterns in the overall magnetic field. These disturbances can cause flares, sunspots, and more intense solar activity in general. This overlapping of the Sun's magnetic fields is what causes the flipping of the poles to affect the intensity of the Sun. Traditionally, the intensity of the solar cycle, and thus of the magnetic field, is measured using the number of sunspots on the Sun as a proxy; the Sun has the largest number of sunspots at solar maximum.

About a century ago, when instruments started getting more complex, scientists also started using other indicators of magnetic field intensity that are easier to observe or are more correlated with the true intensity. The main alternative index is the 10.7 cm solar flux, which is a measure of how much energy the Sun emits at the 10.7 cm wavelength; this measure is used because it can be reliably observed regardless of the weather or many other factors (F10.7 Cm Radio, n.d.). As the intensity of the Sun's magnetic field increases, there are more solar flares and CMEs. While there are always maximums and minimums, the intensity of each solar cycle varies, with some cycles being much more or much less intense than others.

What's going on with Solar Cycle 25?

Before every solar cycle, there are scientific panels that predict its intensity. For Solar Cycle 25, which began in late 2019, these panels predicted something completely different than what has actually happened: "that Solar Cycle 25, following a relatively weak Solar Cycle 24, would also be weak" (NOAA Forecast, 2023). Instead of being weak, however, Solar Cycle 25 is nearly double its expected strength and is one of the most powerful modern cycles. In August of 2024, the Sun had 215.5 sunspots compared to the only 107.8 predicted (Solar Cycle, n.d.). This disparity is also present in solar flux numbers, with the solar flux in August 2024 being 245.6 solar flux units instead of the predicted 130.6. Since the strength of Solar Cycle 25 is so much higher than anticipated, preparations by governments and industries based on the original estimates could be in jeopardy. This solar cycle exemplifies how uncertain the scientific community still is when it comes to predicting solar activity in future solar cycles.

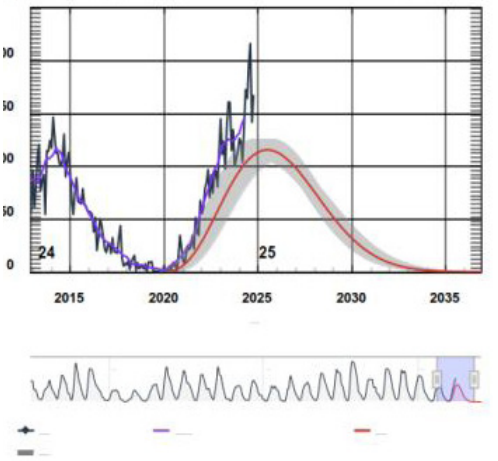


Fig. 1: A graph of the actual smoothed monthly sunspot number (in blue) versus the prediction (in red). (Solar Cycle, n.d.)

Methods of Predicting Solar Cycles

There are three main methods for predicting solar cycles: extrapolation methods, model-based methods, and precursor methods (Petrovay, 2010). These methods utilize different approaches for predicting the intensity of a solar cycle. Extrapolation (also called climatological) methods use mathematics and statistics to forecast future solar cycles by examining trends in data of many previous cycles and trying to find patterns (Petrovay, 2010). The reliability and accuracy of this technique is improving as of late, especially with the increased data and computational power available to scientists. For example, some new developments in the field involve using artificial intelligence to find better statistical analysis methods (Pesnell, 2008). Another variation of this method, recent climatology, uses statistical analysis only of recent solar cycles to predict the potential sunspot magnitude. Finally, precursor methods look at a specific characteristic of the Sun during the previous solar cycle and then use that to predict the intensity of the next solar cycle; precursor methods are the main and the most commonly used type of forecast (Petrovay 2010). One characteristic of the Sun on which predictions are often based is the solar polar magnetic field during the minimum of

a cycle (Kumar et al., 2021). This method predicted Solar Cycle 24 correctly and is one of the prediction methods best-regarded by the scientific community at large. Another popular indicator for precursor methods to be based off of is the geomagnetic activity at solar minimum (Petrovay 2010). Precursor models based on different indicators often produce differing predictions. While prediction methods other than those expounded upon above, like spectral forecast methods, do exist (Pesnell 2008), these three are currently the major categories of prediction methods (Petrovay 2010). Although some combination of these prediction methods normally lead to a reasonably accurate future estimate, for Solar Cycle 25 almost all of these predictions were completely off. This drastic error in prediction shows how much is still not understood about the Sun, and how new methods or variations on current methods are needed.

A New Method

A paper published by Scott McIntosh and four others broke from other predictions of Solar Cycle 25, and said that it "could be among the strongest sunspot cycles ever observed … and it is highly likely that it will certainly be stronger than present SC24 [Solar Cycle 24]" (McIntosh et al. 2020). They had 68% confidence that the maximum sunspot number of Solar Cycle 25 would be between 204 and 254. Despite their model giving a very different result than almost all of the other models, as the peak of the solar cycle is nearing their model is being proved correct; the highest (monthly smoothed) sunspot number measured so far this cycle is 215.5. This was measured in August of 2024, near the peak of the solar cycle. The paper based this unorthodox prediction of the solar cycle on a theory formulated by McIntosh and some of his co-authors in an earlier paper; that the intensity of solar cycles can be predicted by the movement and timing of magnetic bands in the Sun (McIntosh 2014). According to the theory, as the Hale cycle progresses so too do sets of magnetic bands in their course; partway into each solar cycle, the set of bands for the next solar cycle is generated at about ±55° in latitude and then move from the poles to the equator over the course of 19 years (Dikpati et al. 2019). This process is shown in Fig. 2. When the bands of the Sun's magnetic field meet at the equator, they cancel out in a termination event. This termination event is what causes the switch from one solar cycle to another.

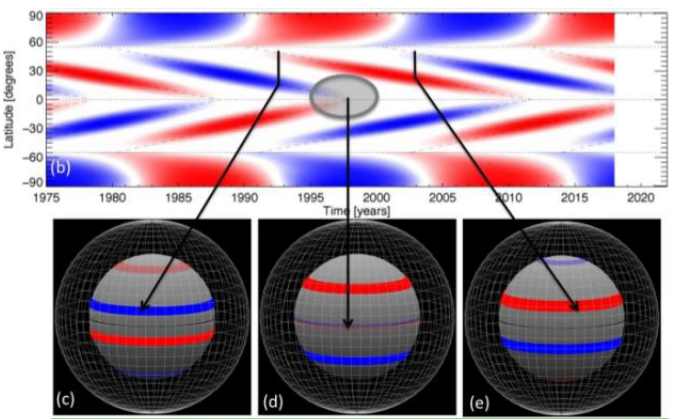


Fig. 2: Diagram of the latitude of the magnetic bands on the Sun over time (Dikpati et al., 2019). As time passes, the bands get closer to the equator, before they collide at the equator and cancel out. The paler/higher up bands are the bands that correspond to the next solar cycle.

The canceling out of magnetic fields causes solar cycles by catalyzing the formation of a "solar tsunami" (Dikpati et al., 2019). The fluid is released from the hold of the magnetic fields in an event similar to the breaking of a dam when the fields cancel out. When the two groups of fluid collide at the equator, they bounce back and head towards the poles again. This tsunami of fluid then causes sunspots at higher latitudes and kickstarts the next solar cycle, starting off with weaker sunspots in the higher latitudes and then moving down to the equator and increasing in strength. A correlation was also found between the intensity of the next solar cycle and the separation (in time) of the two

terminator events that preceded that cycle (McIntosh et al., 2014). When the time between the two preceding terminators was the greatest, solar cycles were weakest, and vice versa. As the terminator event preceding each solar cycle happens near the start of that solar cycle, if this method is proved correct it would be able to predict the intensity of cycles accurately at or near their beginning. It is a new and potentially more precise method for forecasting solar cycles; this comes at a time when the accuracy of solar cycle forecasts is only increasing in importance as the world becomes more technological and prone to disruption by solar activity.

The Importance of Predicting Solar Cycles Accurately

Increased solar activity brings more solar flares, CMEs, and solar radiation. Aside from the danger posed by an incredibly strong but unlikely solar flare/CME like the Carrington event, which could destroy huge amounts of Earth's infrastructure, milder solar flares can pose dangers to orbiting spacecraft, power grids, and radio signals (Fox, 2022). For example, solar flares can cause an unexpected source of drag for spacecraft, which can cause them to deorbit or diverge from their intended orbits (Hathaway, 2015). One of the reasons Skylab, the United States' first space station, deorbited earlier than anticipated was increased atmospheric drag due to higher than expected solar activity. Another more recent example was in February of 2022, when a solar storm caused 40 freshly-launched Starlink satellites to reenter the Earth's atmosphere soon after launch because of an increase in drag (Howell 2024). Another storm in May of 2024 caused GPS outages for farming equipment and other systems. Stronger flares can destroy or damage equipment related to power lines and pipelines. In fields of study and industries where flares or changes in solar activity can make a large difference, accurate solar cycle predictions are crucial. Better prediction methods will help improve certainty in fields dependent on solar activity and improve the reliability of systems that are essential to our current way of life.

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Digital Doctor: How AI is Revolutionizing Medical Diagnostics

Natalia Garment, '26

What are medical diagnostics?

Medical diagnostics are the foundation of modern healthcare. It involves identifying the cause of a patient's symptoms and determining the most appropriate treatment. Traditionally, this process relies on blood work, imaging, biopsies, and the physician's ability to monitor and synthesize this information into a concrete diagnosis and treatment plan. However, recent advancements in artificial intelligence (AI) are offering an opportunity to increase the efficiency and accuracy of this process (Al-Antari, 2023). Though integrating this technology may streamline work and reduce costs, it raises many ethical and practical challenges.

The Role of AI in Diagnostics

AI can be applied to diagnostics through the use of its machine learning models. A machine learning model is an algorithm that utilizes datasets to predict the class a data instance belongs to. (Gurevich et al., 2022) For example, using data from past diagnosis to associate a person's symptoms with a condition. Additionally, AI systems are capable of processing large amounts of visual data with speed and precision, enabling early detection of diseases. In medical imaging, AI can aid radiologists in identifying signs and patterns in radiographs, MRIs, and CT scans such as tumors, fractures, or infections, reducing the risk of human error (Spectral AI, 2024). Making the diagnostic process more efficient and accessible has the potential to aid doctors and patients alike, especially in underserved areas with limited healthcare personnel. This is especially applicable in oncology where time is of the essence and waiting for overt symptoms to show in the advanced stages isn't an option. One tech giant, Google, is developing a tool to help individuals researching conditions. They demonstrated the potential of their model to analyze a collection of mammograms with the same or better accuracy than oncologists (Google Health, 2024). They are now working to see how

the tool may be adopted in clinical practice to more readily diagnose early signs of disease and allow timely interventions that can improve patient outcomes and reduce treatment costs. Another critical area where AI is making strides is personalized medicine. By analyzing patient data (ex. medical history, genetics, and lifestyle factors), AI can help create treatment plans unique to each patient (Spectral AI, 2024). This approach moves away from the typical "one-size-fits-all" treatment model that is easier for doctors to administer. This could benefit people with chronic illnesses by adjusting their treatment plans to suit their needs.

The Economics of AI Diagnostics

AI's impact as a diagnostic tool extends beyond the healthcare system. The potential for AI integration to reduce costs is a huge motivator for adoption. A report from the McKinsey management consulting company estimates that AI could save the U.S. healthcare system between $200 and $360 billion dollars annually (5-10%) of total healthcare spending (Sahni et al., 2023). These savings span across hospital domains as shown in Figure 1.

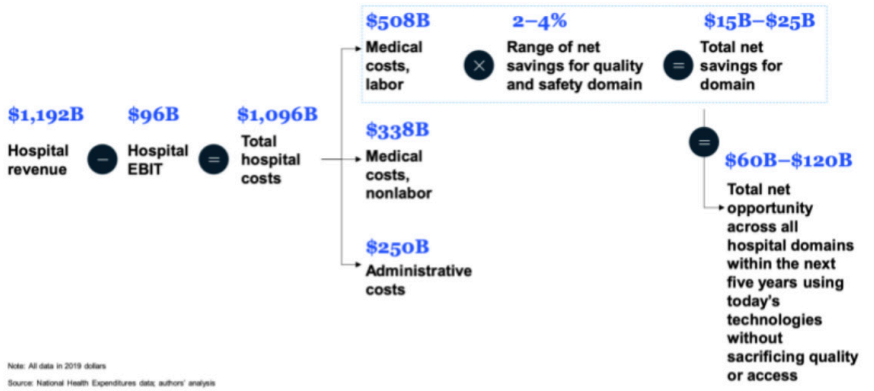


Figure 1: Potential savings across hospital domains using new technology

The Dilemma with AI

Despite promises of AI revolutionizing medicine, several issues remain before it can be implemented. AI's machine learning mechanism is new and with that comes the hurdles of training it with the requisite data. Bias in AI algorithms occurs when AI systems are trained on general datasets that don't represent specific populations (Al-Antari, 2023). This has already raised concern amongst the general public, but it is truly imperative to address when initially considering using these models in healthcare, as a biased algorithm could lead to misdiagnosis. Unfortunately, the issue becomes even more complicated when trying to broaden AI's dataset. Regulation of patient privacy, data security, and transparency from AI companies is crucial to establishing clear ethical guidelines (Al-Antari, 2023). As it currently stands, health information given to third parties is unregulated by the Health Insurance Portability and Accountability Act (HIPAA). In other words, if one consents to give their information to a third party (such as an AI company) that information is not protected under the non-disclosure regulations in place by HIPAA. Luckily, as the technology progresses we are more likely to see guidelines on data established, diminishing problems such as security and bias.

Takeaway

AI has the potential to revolutionize medical diagnostics, offering the promise of accurate, efficient,

and personalized care. However, challenges must be addressed before we see any widespread integration into the healthcare system.

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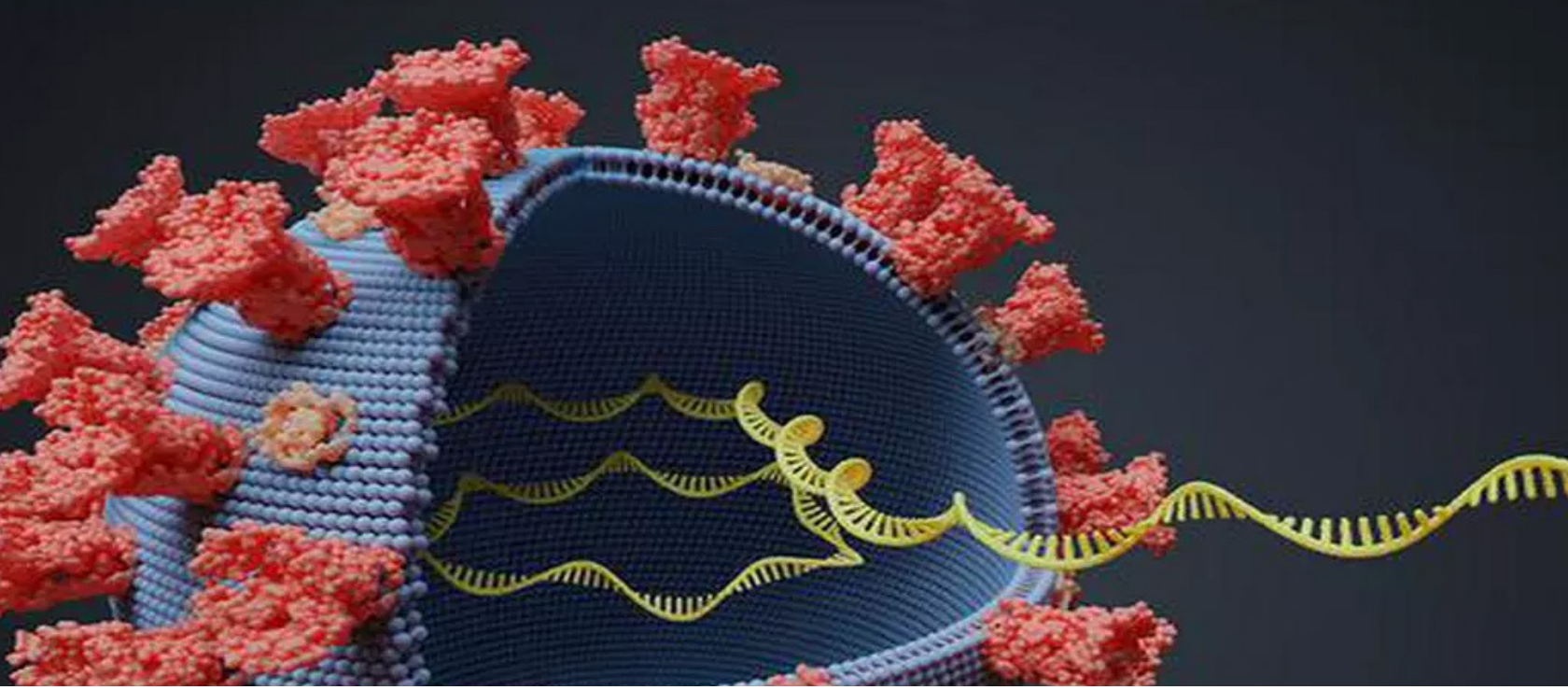
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LucaProt: Discovery of Hidden Viruses

Olivia Morgeson, '26

RNA Virsuses and Previous Studies

RNA viruses infect a wide variety of species, including humans. An RNA virus is a virus that has ribonucleic acid (RNA) as its genetic material, rather than deoxyribonucleic acid (DNA). Due to the error rate of the enzymes involved in RNA replication, RNA viruses exhibit significantly higher mutation rates than DNA viruses. This results in a wide variety of RNA virus variants that rapidly evolve and adapt to new hosts, which makes creating treatments and vaccines for the viruses very challenging; yet, RNA viruses serve a crucial role as major components in global ecosystems. Their importance has recently garnered attention through large-scale virus discovery initiatives in animals, plants, fungi, aquatic environments, marine environments, soil environments, and planetary metra-transcriptomes. These studies relied on the analysis of RNA-dependent RNA polymerase (RdRP) sequences, RdRP being the enzyme that catalyzes the replication process of the RNA in the RNA virus. This has resulted in the discovery of tens of thousands of new virus species, significantly expanding the virosphere, the known world of viruses. However, these discoveries still had limitations which left unknowns; the sequence-similarity searching methods used to detect the viruses were unable to identify highly divergent RdRP sequences. Additionally, the profile-alignment based approach had a high false negative rate, missing a significant amount of viruses. The viruses not found create what is known as the dark matter of the virosphere, the viruses containing divergent RdRPs that are extremely challenging to detect. Uncovering this "dark matter" seemed unlikely, given the limited current methods and tools; that is, until the development of a new approach.

The Development of LucaProt and Expansion of the Virosphere

A research team at Sun Yat-sen University developed the tool LucaProt, an AI model based on transformer architecture which enables better detection of the highly divergent RNA viruses. Sequencing data and ESMFold protein prediction data were incorporated into the model, along with the model being trained to recognize viral RdRPs. This method resulted in the discovery of 513,134

RNA viral contigs, representing 161,979 potential viral species and 180 RNA viral supergroups. Additionally, LucaProt identified 70,458 unique viruses, including 60 previously unidentified supergroups. Figure A visualizes LucaProt's discoveries' contributions to the virosphere. LucaProt also identified several extremely long and complex RNA virus genomes, including one that was 47.3 kilobases long, which exceeds the longest RNA virus genome previously identified. Identified viruses also were found in extreme environments including hot springs, hydrothermal vents, and salt lakes. Along with making numerous breakthrough discoveries, LucaProt also proved to be an extraordinary new model. It achieved the highest recall rate while maintaining a low false positive rate, as well as demonstrating reasonable computational efficiency. Moreover, LucaProt outperformed other methods in terms of precision and long-sequence processing. The development of LucaProt and its findings solved the previous problem of undetectable, "dark matter" viruses, furthering the diversity of known viruses and providing a new direction for research on the virosphere.

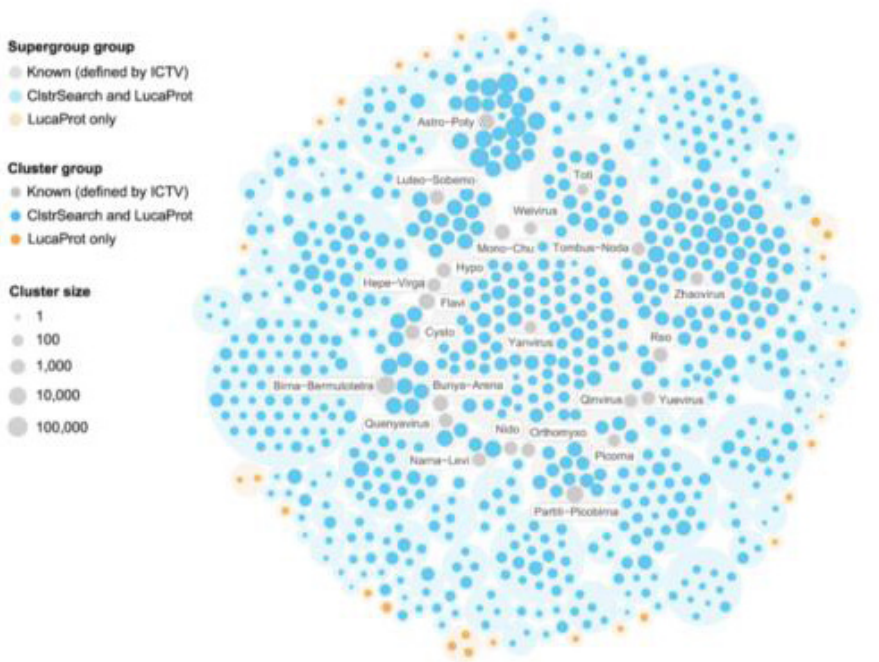


Figure A: Map displaying LucaProt's contributions to the expansion of the virosphere

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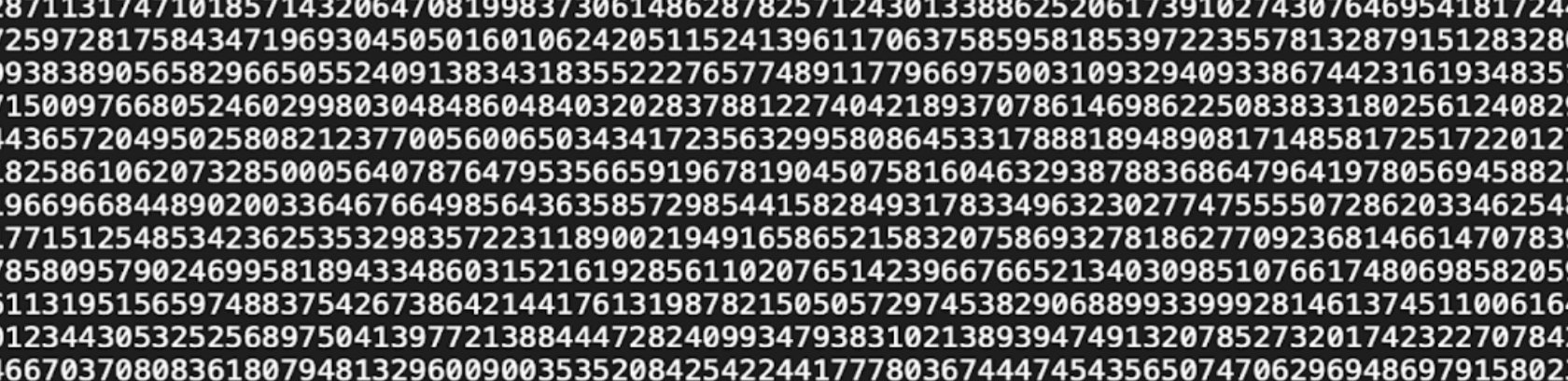
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GIMPS and the Search for New Prime Numbers

Ayush Rudra, '27

On October 11, 2024, Luke Durant was packing his suitcase before a trip to his home in Alabama when he decided to check his computer, which was running a prime checking program known as GIMPS. He figured, like all the other times he had checked in the last year, that he hadn't found anything. But, this time, he was wrong. This time, he had discovered the largest prime known to man. At 41,024,230 digits, this new prime, denoted as M136,279,841 is the largest prime (and largest Mersenne prime) discovered in around six years. It has 16 million more digits than the previous largest prime. The notation arises from the value of the prime being 2136,279,841 - 1 (Brasch, 2024).

Mersenne Primes

Recall that prime numbers are numbers that are only divisible by 1 and themselves. Some examples include 2, 3, 5, and 7. Numbers that are not prime are called composite. 1 is generally considered to be neither prime nor composite. Mersenne numbers are numbers of the form 2n - 1, first studied by French mathematician Marin Mersenne (1588-1648). Similarly, Mersenne primes are any primes of the form 2n - 1. For example, 31 is a Mersenne prime because 31 = 25 - 1. Another fact about Mersenne primes is that the exponent of the 2 is always prime. So, because 31 = 25 - 1 is prime, that means that 5 is prime. It is important to not mix this up with the converse. In other words, just because a prime p is prime, that does not mean that 2p - 1 is prime. For example, 11 is prime, but 211 - 1 = 2047 is not prime, as it can be factored into 23 and 89. Mersenne primes lend themselves nicely to all things computer science-related, as their binary form is simply repeated 1s. There are still many things unknown about Mersenne primes: the biggest looming question is whether there are a finite number of them.

What is GIMPS?

Short for Great Internet Mersenne Prime Search, GIMPS is a program in which volunteers download

free software onto their computers. The software constantly runs in the background, making use of extra computer processing power to search for more Mersenne primes. The search started in 1996 with George Woltman, and discovered its first new prime in November of that year. In fact, since its creation, the GIMPS program is responsible for discovering the most recent 18 of the total 52 Mersenne primes currently known.

How does the GIMPS program search for primes?

Until 2018, GIMPS used the Lucas-Lehmer primality test to test for primes. It was first discovered by Édouard Lucas in 1878 and then rigorously proved by Derrick Lehmer in 1930. This is how the test works. Consider the sequence of numbers 4, 14, 194, 37634, and so on. In general, s0 = 4 and the sn is equal to (sn-1)2 - 2. The Lucas-Lehmer test states that a Mersenne number Mp is prime if and only if sp-2 is divisible by Mp. For example, s5-2 = s3 = 37634 is divisible by M5 = 25 - 1 = 31. So, M5 = 31 is a Mersenne prime ("Lucas-Lehmer Test"). Because of its complexity, there will not be a proof included here. Obviously, this method is useless for humans to use due to the magnitude of the numbers involved. However, due to the inherently binary structure of Mersenne numbers, it makes it much easier for computers. Still, the Lucas-Lehmer method has one large problem: all results have to be double checked. So, in 2018, GIMPS switched to a different method of prime verification called probable primes, or PRP for short ("Mathematics and Research Strategy"). The concept of PRP testing is actually relatively straightforward. It relies on Fermat's little theorem, a not-so-little result in number theory. This states that given a prime p, then for all values of n that are not divisible by p, np-1 - 1 is divisible by p ("Fermat's Little Theorem". The idea is that if we are testing if some number p0 is prime, we can select random values of n and check if Fermat's theorem holds. If it holds for one or more values of n, p0 is probably a prime (hence the name). For example, since 38220 - 1 is divisible by 221, 221 would pass the PRP test and would likely be a prime. However, it would be a false positive because 24220 - 1 is not divisible by 221. Indeed, 221 = 13 • 17. The great benefit of PRP is that there are no false negatives, meaning every number that fails the test is for sure not a prime (this follows from the contrapositive of Fermat's theorem). Additionally, it is much faster at generating a list of potential primes, with the caveat of there being a small chance of false positives. However, the amount of numbers that pass the PRP test are so small that it is worthwhile to use the more time consuming and accurate tests on this small subset of numbers.

What is so special about Durant's number?

The discovery of this new prime marks the end of an era. Before Durant, all primes discovered under the GIMPS program were discovered by ordinary personal computers using a central processing unit (CPU). However, Durant's prime was found via cloud computing, relying on a graphics processing unit (GPU). His "computer" was made of thousands of GPUs spanning over 17 different countries (Paul, 2024). The main difference between CPU and GPU is that while CPUs can execute many different directions, GPUs can take one set of directions and can do them extremely fast. This makes GPUs better for tasks like GIMPS, which involve brute force computation (What's the Difference Between GPUs and CPUs?). After a year of running Durant's computer system, the new prime was discovered and verified.

Applications

Mersenne primes, and primes in general, have many applications in computer science and adjacent fields. For example, they are used for making pseudorandom numbers. Mersenne Twister is an algorithm that generates extremely long random-looking numbers. It has a period of M19937, which means that it can make sequences of that length before repeating itself (Hwu). The structure and

size of Mersenne primes make it optimal for things like this. Primes also have use in everyday life, like encryption. The RSA encryption algorithm, said to secure 90% of the Internet, uses the fact that finding the prime factors of extremely large numbers is difficult to do. In general, the randomness of primes, especially large primes, helps for all things cyber-security related, like credit card fraud, internet security, and cryptography ("Prime Numbers Applications in Real Life").

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Can Burying Wood Help Fight Climate Change?

Sienna Schwartz, '27

In 2013 Ning Zeng, a climate scientist at the University of Maryland, and his colleagues were digging a trench in the Canadian province of Quebec. They were planning on burying 35 tons of wood, covered with clay soil, for 9 years. During excavation, they came across this very old, yet incredibly important log, see Figure 1.



Figure 1: The 3,775 Year-Old Log Unearthed in Quebec

Caption: Scientists unearthed this 3,775 year-old piece of wood that retained at least 95 percent of its carbon, thanks to being covered by a layer of clay soil (Zeng et al. 2024)

The Development of LucaProt and Expansion of the Virosphere

"I remember standing there just staring at it," Zeng recalls. He remembers thinking, "Wow, do we really need to continue our experiment? The evidence is already here, and better than we could do."

The log came from an Eastern red cedar tree that lived thousands of years ago. It was found underneath two meters of clay soil and retained 95% of its carbon. Zeng suspects that the layer of clay soil between the buried wood and the atmosphere helped to prevent any oxygen from reaching the log. Zeng's insights show that burying wood is a viable option for fighting climate change when a new solution is desperately needed.

According to the Intergovernmental Panel on Climate Change, "curbing greenhouse gas emissions isn't enough to meet global climate targets." Annual CO2 emissions have increased since the start of the Industrial Revolution in 1750. According to the US Global Change Research Program, "Global monthly average concentrations of carbon dioxide have risen steadily from 337 parts per million in 1979 (averaged over the year) to 417 parts per million in 2022, an increase of more than 20% in 44 years," see Figure 2.

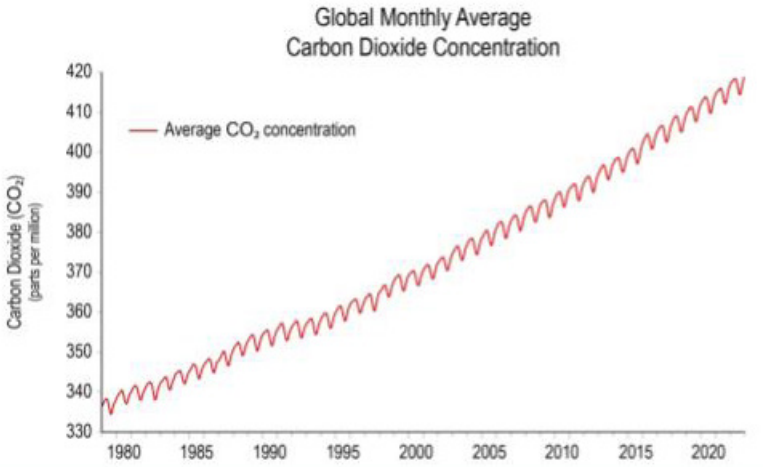


Figure 2: Global Monthly Average for CO2 Increases Since the Data Collection Began

Concentrations of carbon dioxide have risen steadily from 1979 to 2022 (Global Monthly Average Carbon Dioxide Concentration, 2022).

Globally, forests absorb nearly 16 billion metric tonnes of carbon dioxide per year, and currently hold 861 gigatonnes of carbon in their branches, leaves, roots, and soils. When a tree dies, it releases all of its stored carbon back into the atmosphere, further contributing to carbon emissions. Preventing just a fraction of that decomposition by burying wood could help reach the global goal of reducing carbon dioxide in the atmosphere, see Figure 3.

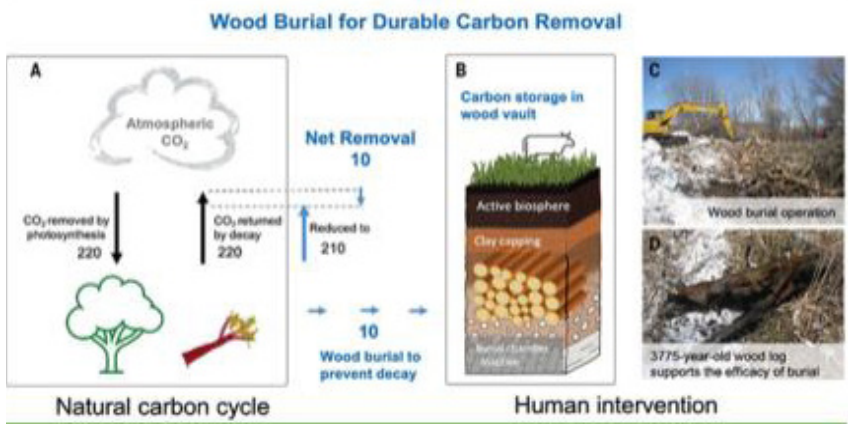


Figure 3: The Wood Burial Study

This wood burial study found that burial in clay reduces decomposition and carbon release (Climate.gov Staff, 2024)

Researchers estimate that burying wood could cost $30 to $100 dollars per ton of carbon dioxide. Since it is such a cost effective solution, many are pushing to implement it into today's society. Buried biomass from discarded wood could isolate up to 10 gigatons of carbon per year.
Daniel Sanchez—an environmental scientist at the University of California, Berkeley—sums up the possible future effects of wood burying by stating, "High-durability, low-cost climate solutions like these hold immense promise for fighting climate change.

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K2P Channels: The Secret to Health and Sleep?

Eric Wilson, '26

K2P channels, or potassium ion gated channels, are pathways located in the cellular membrane, the layer of phospholipids surrounding a cell. K2P channels regulate the amount of potassium (K+) that goes through the cell membrane, allowing a cell to control the amount of potassium inside of it at all times. K2P channels are significant in humans, and channel abnormalities have been linked to human disease. Malfunctioning or mutated K2P channels can cause terminal heart conditions, such as raising cardiac action, which can increase the chance of cardiac arrest. K2P channels are traditionally studied through mouse DNA, as the K2P channels in mice most directly mirror those in humans. There are 15 types of these channels expressed in humans, however for simplicity research focuses on TASK-1 K2P channels (Figure 1).

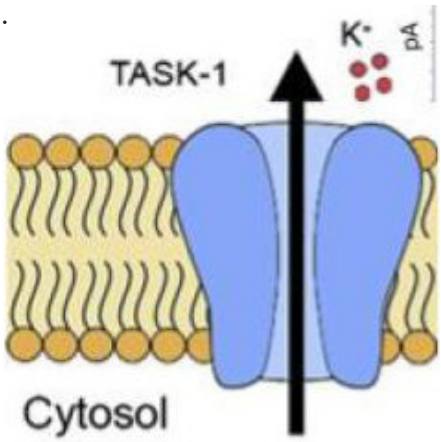


Figure 1: Task-1 K2P Channel

How do they work?

TASK-1 K2P channels have "caps," similar in shape and function to a cork, which block the opening of the channel once the cell reaches an optimum potassium concentration. These caps are not fully understood, however it is speculated that a triggering mechanism inside of the channel itself, in response to an optimum potassium gradient inside of the cell, sends a message to the cap to close

the channel. When the channel closes, it prevents potassium from entering the cell. In cells with malfunctioning K2P channels, these caps do not function properly. The cap malfunction alters the potassium gradient inside of the cell, causing problems for the organism.

Why are they important?

K2P channels serve a variety of functions within the body. One such function is to regulate the amount of potassium within a cell. This is important, because an influx of potassium into a cell can cause the cell to denature, throwing off the balance of potassium within the cell, and in most cases kill the cell. K2P channels also have stark impacts on cardiac action. Cardiac action is the electrical event responsible for the generation of the cardiac impulse and conduction of the electrical impulse through the myocardium and specialized cardiac conduction system. This system is responsible for contracting the walls of the heart, allowing it to pump blood throughout the body. If K2P channels are malfunctioning, this contractile system does not function properly. This malfunction heightens the chance for a cardiac arrest, which can be fatal, especially for people of old age.

Role in Anesthesia and Sleep

K2P channels have also been linked to anesthetics. It has recently been proven that these channels open and close in response to anesthetics, leading to the conclusion that these channels may aid the body into reaching a state of unconsciousness. This is due to these channels having control over the contractile walls of the heart. Under anesthesia, it has been noted that heart rate slows down significantly.

Breakthroughs for Pharmaceutical Development

Recently, a team of researchers working for the Kavli Institute for Nanoscience Discovery at the University of Oxford used a technique called CryoEM, "a method used to image frozen biological molecules without the use of structure-altering dyes or fixatives or the need for crystallization," (National Institute of Health) to map the structure of the TASK-1 K2P channel. This team of researchers found the optimal pH that the TASK-1 channel opens, and also confirmed that previous imaging of this channel, which had used outdated techniques, was accurate. This development supports and encourages new research to continue for a K2P channel drug, which could be used to manually open and close these important channels. Anesthesiologists could use the new drug to control a patient's heart contraction rate, while under anesthesia, which would make it safer in the operating room. Development of a new drug targeting these channels would also help prevent or reduce the risk of cardiac arrest in humans, saving countless lives.

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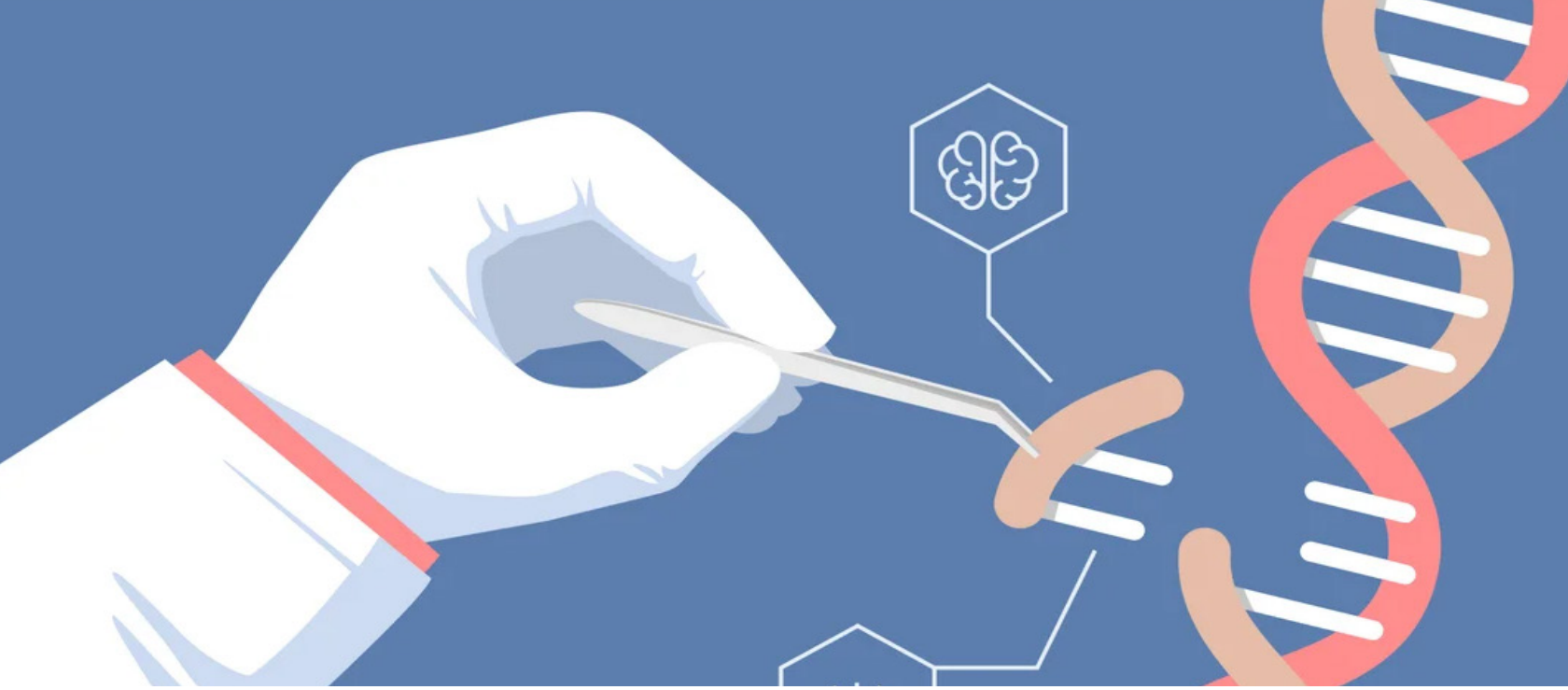
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CRISPR's Transformation from Tool to Medicine

Jessie Zhang, '26

What is CRISPR?

Clustered Regularly Interspaced Short Palindromic Repeats, or more formally known as CRISPR, is a form of gene-editing technology that allows scientists to make precise modifications to an organism's DNA. The Casfamily of enzymes were originally discovered in E. coli as a defense mechanism against viruses, but now serve as a tool that grants researchers the ability to either deactivate a gene, correct mutations, or insert new genetic material. Thus, companies such as Vertex Pharmaceuticals Incorporated have expertly manipulated this tool, creating such inventions like CASGEVY.

Previous Advancements

First developed in 2012 by scientists Jennifer Doudna and Emmanuelle Charpentier, CRISPR-Cas9 —the most widely used enzyme of CRISPR— employs an RNA guide to specify specific DNA sequences, cutting the DNA at the desired location.

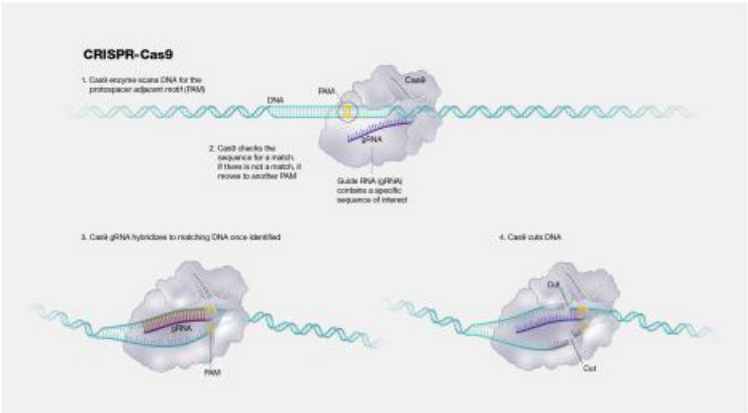


Figure 1: Visual of the CRISPR gene editing process.

One of CRISPR-Cas9's successes was its ability to help researchers better understand how specific genes influence biological processes and diseases. By "knocking out" or inserting genes, scientists have been able to model human diseases in animals, providing new and valuable insights into the genetic basis of conditions such as cancer, Alzheimer's disease, and heart disease. It is uses like this that have made CRISPR a profound innovation on genetic studies, offering a more accurate alternative to prior gene-editing methods.

A New Role

Initially, CRISPR was confined to the lab due to its unpredictability, where it revolutionized research in genetics, molecular biology, and even biotechnology. However, its potential for treating genetic disorders, enhancing crops, and even combating infectious diseases propelled CRISPR into real-world applications. On Friday, December 8 of 2023, the FDA approved CASGEVY (exagamglogene autotemcel) for treatment of sickle cell disease in patients 12 years and older with recurrent vaso-occlusive crises. CASGEVY is the first therapy approved for use in the United States that uses CRISPR gene-editing technology to edit a patient's gene.



Dr. Monica Bhatia, a pediatric hematologist-oncologist and director of the Pediatric Stem Cell Transplant Program at NewYork-Presbyterian Morgan Stanley Children's Hospital.

According to hematologist-oncologist Monica Bhatia, MD, associate professor of pediatrics at Columbia University Vagelos College of Physicians and Surgeons, who has treated patients with the therapy in the clinical trial that led to approval, "by editing certain cells, we decrease pain caused by vaso-occlusive crises and decrease hospitalizations for people with sickle cell disease." Outside the body, CRISPR technology is used to make a precise edit to a gene that regulates the production of fetal hemoglobin, a form of hemoglobin normally produced in the fetus but typically switches off shortly after birth. After CASGEVY is infused into the patient's own blood stem cells, the edited stem cells engraft in the bone marrow, where they begin to produce red blood cells that contain fetal hemoglobin. Once these red blood cells reach the sufficient and desired level of fetal hemoglobin, they no longer sickle or become sticky, preventing the complications associated with sickle cell disease. It's a completely new concept in which scientists are reprogramming the patient's own genes.

Potential for Long-term Cure

In the short term, it can not be determined if CASGEVY will be a sustainable cure to sickle cell disease yet. Despite this, in the study, people who were treated with CRISPR gene therapy have been free from vaso-occlusive crises so far. Although, because the treatment is so new, scientists

must wait to see if the results last. They will need to continue to follow these patients to see if the complications of sickle cell disease—organ damage and strokes, in particular—are prevented by the CRISPR treatment. Due to this, doctors will follow the patients for the next 15 years to learn more.

The Future

The early results from this study have shown encouraging success, with many of the patients on trial experiencing reduced symptoms and improved overall health. With this information in hand, it is clear that CRISPR offers a transformative potential for those suffering from sickle cell disease, while even extending beyond just sickle cell disease. CRISPR technology in medicine might be especially important at a time when conventional treatments are facing limitations. For example, antibiotics are becoming less effective due to the rise of antibiotic resistance. Many bacteria have evolved to resist the drugs that once killed them, rendering some infections nearly impossible to treat with traditional antibiotics. Tools like CRISPR offer a promising alternative to tackling diseases at their genetic root. As CRISPR based therapies continue to evolve, they could lead to improved treatments for people suffering from genetic diseases, with the means to change the entire landscape of medicine.

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Porphyrion; Giant Astrophysical Jets

Zachary Gottlieb, ‘25

In September, a group of physicists published their findings on what is now the largest known pair of black hole jets in the universe—Porphyrion, named after the giant from Greek mythology (Oei et al., 2024). In a galaxy 10 times more massive than our own, Porphyrion stretches 23 million light-years across, the equivalent length of 140 Milky Ways. Porphyrion’s discovery brings new insights into the physics of black holes and their potential to transport energy and magnetism throughout the cosmos.

Black Hole Mechanisms

At the center of nearly every large galaxy in the universe is a supermassive black hole (SMBH). Though black holes are famous for not letting light escape, they can become the brightest objects in the universe. This phenomenon is called a quasar, an extremely luminous type of active galactic nuclei (AGN; a galaxy with an SMBH actively “accreting” matter). According to processes defined by the mathematician Roy Kerr in the 1960s, as an SMBH spins around its axis, it draws in matter and charged particles which are heated by friction into a plasma. This plasma accretion disk carries with it an electrical current that produces a magnetic field (Orf, 2024). Black hole jets, also known as astrophysical jets, are the consequence of this powerful magnetic field. The Blandford-Znajek process describes the motion of the fields and their influences on the black hole. When the magnetic field lines fall onto the black hole, its rotation twists the field into a helix along its axis. Just on Earth as moving magnetic fields generate voltage, the voltage created by an SMBH is powerful enough to draw up and eject some of the charged particles in two galaxy-sized jets (Wolchover, 2021). Coincidently, the particles funneled into the jets are accelerated to speeds close to thespeed of light. When looking through radio telescopes on Earth, these particles are observed as synchrotron radiation, visible across a wide range of wavelengths. The first recorded image of a black hole, the SMBH at the center of the Messier 87 galaxy, confirmed the presence of the warping magnetic field and further solidified the existence of astrophysical jets (Klesman, 2023).

Finding Porphyrion

The discovery of Porphyrion happened as an accident. Back in 2018, Marjin Oei, the lead author of the study, was initially searching for a radio signature of the cosmic web. The cosmic web is what defines the large-scale structure of the universe—it is a network of voids and filaments of matter that connect the galaxies to one another. Using Europe's International LOFAR Telescope (ILT), Oei and his team incidentally found thousands of new streaks that seemed to originate from nearly every large galaxy with an AGN (Clavin, 2024). By using machine learning, and with the help of citizen scientists, the team pored over radio imagings to search for more hidden jets, increasing "the number of known mpc-scale outflows from a few hundred to more than 11,000" (Oei et al., 2024, p. 537). Porphyrion was the largest. Bordering the detection limits of current leading telescopes, Porphyrion is 7.5 billion years from Earth and seen at the age of 6.3 billion years after the Big Bang. About 7 mpc in length, for Porphyrion to be so huge, the "black hole responsible would have needed to ingest about a sun's worth of matter each year for a billion years" (Wilkins, 2024). To make and confirm these measurements, the team initially used ILT to capture the radio image due to its sensitivity to synchrotron radiation. Once Porphyrion was identified, they used the upgraded Giant Metrewave Radio Telescope (uGMRT) to determine the host galaxy. In measuring spectroscopic redshift and other factors that could influence its distance measurement, the team used the W. M. Keck Observatory in Hawaii. After accounting for all these measurements and data reductions, the team can say with certainty that Porphyrion originates from a cosmic web filament (Lea, 2024).

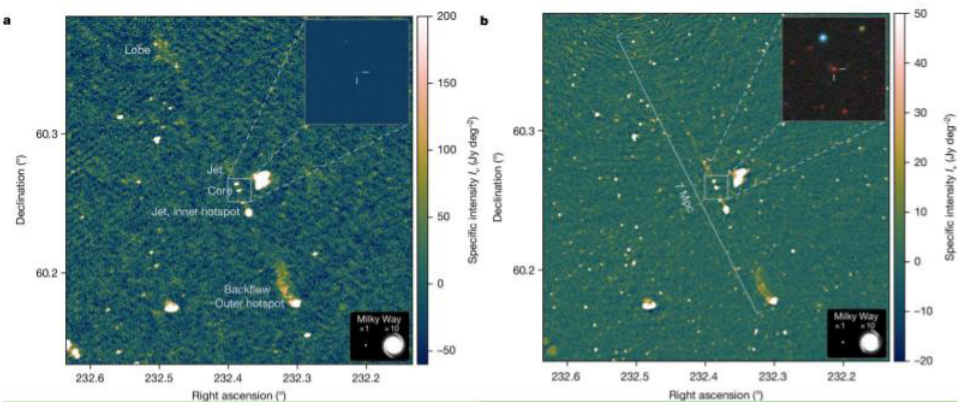


Figure 1: Deep radio images of Porphyrion, a) image taken with ILT, b) image taken with uGMRT. Note the blob of light near Porphyrion's origin is a separate AGN system.

Besides its record-breaking size, the Keck telescope also observed something unexpected about Porphyrion. The observations reveal that Porphyrion emerges from a radiatively efficient AGN (RE AGN; also known as radiative active or quasar mode) as opposed to a radiatively inefficient AGN (RI AGN; jet, kinetic, or radio-jet mode). RE and RI AGN are both feedback mechanisms that influence the accretion disk and thus jet structure. The difference between the two comes from the activity of their host galaxies and the accretion rates of their SMBHs. Generally found in blue star-forming galaxies, RE AGNs "efficiently convert the gravitational potential energy of infalling matter into radiation," whereas RI AGNs are found in massive elliptical galaxies that accrete hot gas at a slower rate (Oei et al., 2024, p. 537). RE AGNs were typical of the early universe, while RI AGNs are more common in the later universe. Additionally, all previously discovered jet outflows of record length were fueled by RI processes (Lea, 2024). It is thus surprising to see Porphyion as a large RE AGN so late in the development of the universe.

Black Hole Mechanisms

In an interview with Caltech, Oei says "If distant jets like these can reach the scale of the cosmic web, then every place in the universe may have been affected by black hole activity at some point

in cosmic time" (Clavin, 2024). It is clear from the study that jets like Porphyrion potentially have a much more significant role in heating and spreading energy and magnetism throughout the universe than previously thought. But it is difficult to measure the extent of this influence. At the present, there are no computers capable of running such simulations. According to Laura Olivera-Nieto from the Max Planck Institute in Germany, "It's truly a challenge to try to understand how this is physically possible. We cannot put it in a computer, it's too big" (Wilkins, 2024). Porphyrion carries energy into the IGM comparable to that released during galaxy cluster mergers. Given the jets are from a younger and denser universe, it's a mystery how Porphyrion is able to extend so far beyond its host galaxy without destabilizing (Orf, 2024). The researchers are also interested in whether these jets have an impact on magnetizing the intergalactic medium (the voids of space that fill the cosmic web), and the magnetism of the universe as a whole. The survey only covered about 15% of the sky, meaning that Porphyrion's discovery is only just the beginning. As better instruments become more widely available, and with the successive development of artificial intelligence and quantum computing, scientists will learn more about these immense jets and their place in the universe.

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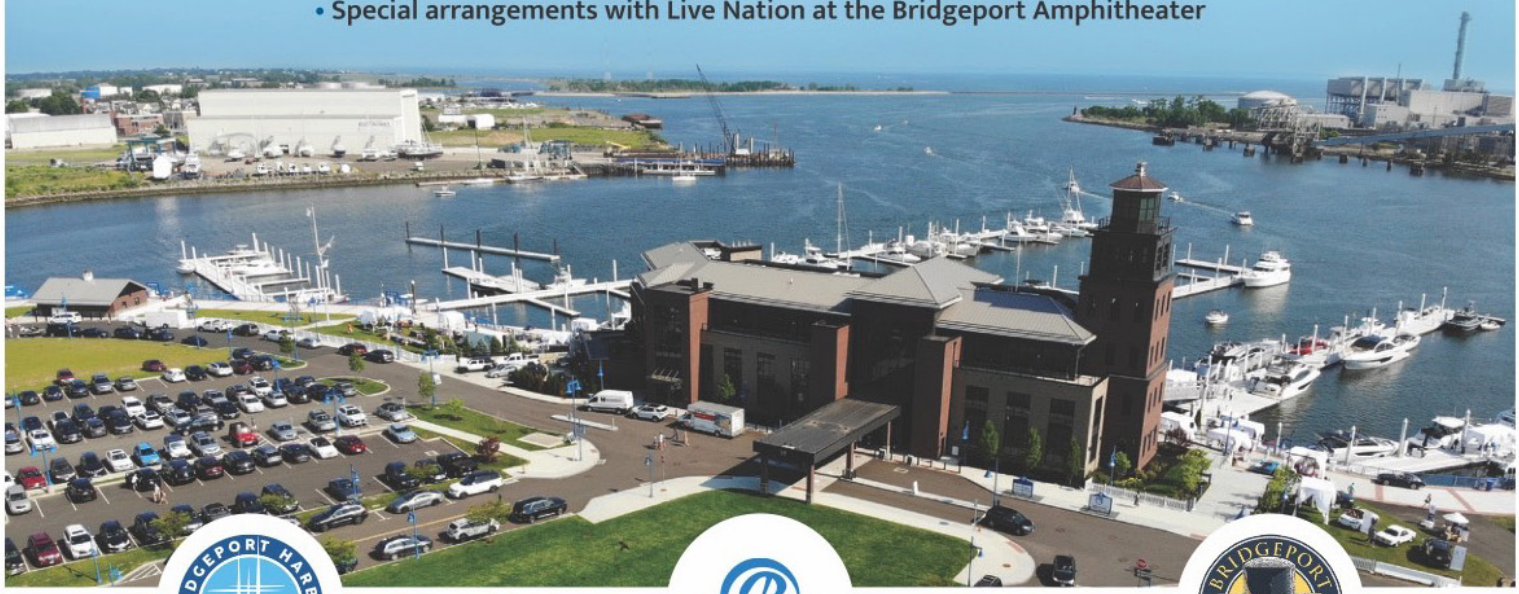



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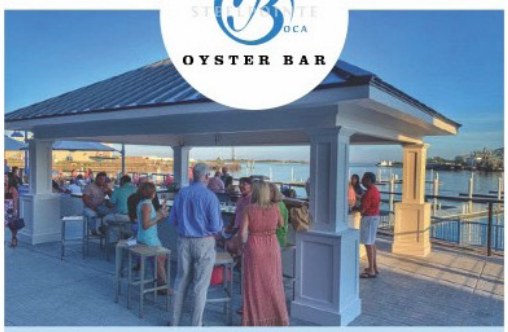
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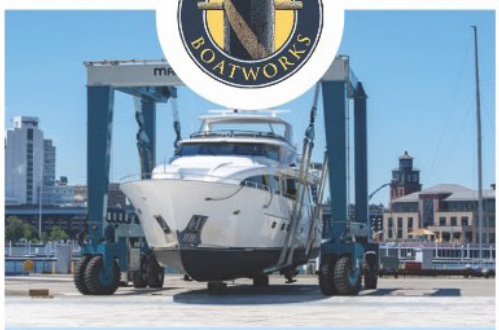




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